Lab 30 — NumPy Random Module for Simulation

Aim

To generate random numbers using NumPy's random module for simulation and sampling tasks.

Algorithm / Procedure

1. Import NumPy.
2. Create a seeded random number generator.
3. Generate uniform random numbers.
4. Generate random integers.
5. Generate normally distributed random samples.
6. Generate random points for Monte Carlo simulation.
7. Count the points inside a unit circle.
8. Estimate π using the Monte Carlo method.
9. Display the results.

Program

```
import numpy as np

rng=np.random.default_rng(seed=42)

print("Uniform random 2x3:\n",np.round(rng.random((2,3)),3))

print("Random integers 0-10:",rng.integers(0,10,size=5))

print("Gaussian samples:",np.round(rng.normal(0,1,size=5),3))

n=100000

points=rng.random((n,2))

inside=np.sum(points[:,0]**2+points[:,1]**2<=1)

print("Estimated pi:",4*inside/n)
```

Output

Uniform random 2x3:

[[0.774 0.439 0.859]

[0.697 0.094 0.976]]

Random integers 0-10: [7 7 7 7 5]

Gaussian samples: [-0.853 0.879 0.778 0.066 1.127]

Estimated pi: 3.15012

Result

NumPy's random module was successfully used to generate reproducible random samples from uniform, integer, and Gaussian distributions. The Monte Carlo simulation also successfully estimated the value of π.